```
clear;
clc;
close all;

%% DESIGN PARAMETERS
Fs = 10000;       % Sampling frequency (Hz)
Ap = 0.5;         % Passband ripple (dB)
As = 30;          % Stopband attenuation (dB)

%% 1. CHEBYSHEV TYPE-I LOW-PASS FILTER DESIGN
fp_lp = 1500;     % Passband edge frequency (Hz)
fs_lp = 2000;     % Stopband edge frequency (Hz)

% Normalize frequencies with respect to Nyquist frequency
wp_lp = fp_lp / (Fs / 2);
ws_lp = fs_lp / (Fs / 2);

% Calculate minimum filter order and cutoff frequency
[N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);

% Design the low-pass filter
[b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');

%% 2. CHEBYSHEV TYPE-I HIGH-PASS FILTER DESIGN
fp_hp = 2500;     % Passband edge frequency (Hz)
fs_hp = 1800;     % Stopband edge frequency (Hz)

% Normalize frequencies with respect to Nyquist frequency
wp_hp = fp_hp / (Fs / 2);
ws_hp = fs_hp / (Fs / 2);

% Calculate minimum filter order and cutoff frequency
[N_hp, Wn_hp] = cheb1ord(wp_hp, ws_hp, Ap, As);

% Design the high-pass filter
[b_hp, a_hp] = cheby1(N_hp, Ap, Wn_hp, 'high');

%% 3. DISPLAY NUMERICAL RESULTS
fprintf('================================================\n');
```

```
================================================
```

```
fprintf(' EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS\n');
```

```
 EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS
```

```
fprintf('================================================\n\n');
```

```
================================================
```

```
% Low-pass filter results
fprintf('--- CHEBYSHEV LOW-PASS FILTER ---\n');
```

```
--- CHEBYSHEV LOW-PASS FILTER ---
```

```
fprintf('Calculated Filter Order (N_lp): %d\n', N_lp);
```

```
Calculated Filter Order (N_lp): 6
```

```
fprintf('Cutoff Frequency (Wn_lp): %.4f Hz\n', ...
    Wn_lp * (Fs / 2));
```

```
Cutoff Frequency (Wn_lp): 1500.0000 Hz
```

```

fprintf('Numerator Coefficients (b_lp):\n');
```

```
Numerator Coefficients (b_lp):
```

```
disp(b_lp);
```

```
    0.0006    0.0038    0.0095    0.0127    0.0095    0.0038    0.0006
```

```

fprintf('Denominator Coefficients (a_lp):\n');
```

```
Denominator Coefficients (a_lp):
```

```
disp(a_lp);
```

```
    1.0000   -3.7650    6.8198   -7.3274    4.8717   -1.8927    0.3368
```

```

% High-pass filter results
fprintf('\n--- CHEBYSHEV HIGH-PASS FILTER ---\n');
```

```
--- CHEBYSHEV HIGH-PASS FILTER ---
```

```
fprintf('Calculated Filter Order (N_hp): %d\n', N_hp);
```

```
Calculated Filter Order (N_hp): 6
```

```
fprintf('Cutoff Frequency (Wn_hp): %.4f Hz\n', ...
    Wn_hp * (Fs / 2));
```

```
Cutoff Frequency (Wn_hp): 2500.0000 Hz
```

```

fprintf('Numerator Coefficients (b_hp):\n');
```

```
Numerator Coefficients (b_hp):
```

```
disp(b_hp);
```

```
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117
```

```

fprintf('Denominator Coefficients (a_hp):\n');
```

Denominator Coefficients (a_hp):

```
disp(a_hp);
```

```
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650
```

```
%% 4. FREQUENCY RESPONSE CALCULATION
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);

% Convert magnitude to decibels
mag_lp_db = 20 * log10(abs(H_lp) + eps);
mag_hp_db = 20 * log10(abs(H_hp) + eps);

% Convert phase to degrees
phase_lp = angle(H_lp) * 180 / pi;
phase_hp = angle(H_hp) * 180 / pi;

%% 5. PLOT FREQUENCY RESPONSES
figure('Name', 'Chebyshev Type-I LPF and HPF Responses', ...
       'NumberTitle', 'off');

% Low-pass filter magnitude response
subplot(2, 2, 1);
plot(f, mag_lp_db, 'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev LPF Magnitude (Order N = ', ...
       num2str(N_lp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);
xlim([0 Fs / 2]);

% Low-pass filter phase response
subplot(2, 2, 2);
plot(f, phase_lp, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev LPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs / 2]);

% High-pass filter magnitude response
subplot(2, 2, 3);
plot(f, mag_hp_db, 'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev HPF Magnitude (Order N = ', ...
       num2str(N_hp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
```

```
ylim([-60 5]);
xlim([0 Fs / 2]);

% High-pass filter phase response
subplot(2, 2, 4);
plot(f, phase_hp, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev HPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs / 2]);

% Improve figure layout
sgtitle('Chebyshev Type-I IIR Filter Analysis');
```

